SMART LIBRARY REQUEST FLOW

ServiceNow Workflow Documentation

Introduction

Libraries have always been one of the most important places for learning — but managing them is a whole different challenge. Keeping track of which books are available, handling borrow requests, and maintaining accurate records can get messy very quickly when it's all done manually. That's the problem this project tries to solve.

- Library is used to manage books and users.
- Many libraries still use manual work, which takes more time.
- So, a Smart Library system is created to make the process easy and automatic.
- ServiceNow, a leading IT Service Management platform, provides the ideal foundation for building this automated solution through its powerful workflow, table, and role-management capabilities.

The Smart Library system uses ServiceNow's Flow Designer to handle the full book request process automatically — right from when a student makes a request, to the librarian approving it, issuing the book, and finally returning it. The goal was to cut down on manual effort, avoid errors, and make the whole process more transparent.

Abstract

This project is about building a Smart Library system using ServiceNow. The idea is simple — when a student wants to borrow a book, they raise a request on the platform. That request then goes to the librarian, who can either approve or reject it using the Flow Designer. Once approved, the book gets issued and the system keeps track of when it needs to be returned. On top of that, the system generates reports that show which books are being borrowed the most.

This document walks through the entire journey of how this system was built — starting from setting up roles and tables, all the way to automating workflows, securing data, and setting up reports. It's meant to give a clear picture of how everything fits together on the ServiceNow platform.

Problem Statement

Before building this system, it was important to understand what was actually going wrong with the existing process. Here are the main issues that came up:

- Library system is mostly manual and takes more time.
- It is difficult to track available and borrowed books.
- Book requests are not managed properly.
- Students cannot easily check book availability.
- Librarians face difficulty in approving and managing requests.
- There is no proper system for tracking and reports.
- Lack of automated notifications means students are unaware of request status changes.
- No centralized dashboard exists for administrators to monitor library usage trends.
- So, a Smart Library system is needed to make the process easy, fast, and accurate.

All of these issues together made the experience frustrating — for students who couldn't easily borrow books, and for librarians who were spending too much time on tasks that could easily be automated.

Objectives

The Smart Library Request Workflow project was designed with the following core objectives to address the identified problems:

- Define roles (student, librarian) to enforce controlled access and ensure each user can only perform actions appropriate to their role.
- Create and configure custom tables for catalog and borrow requests, storing all necessary fields to support the workflow.
- Automate approvals and book issuance using Flow Designer, eliminating the need for manual intervention in routine request processing.

- Apply UI Policies and ACLs to maintain consistency and security across all forms and records.
- Provide reporting capabilities for administrators to track library usage, identify popular books, and monitor borrowing trends.
- Deliver email notifications to students upon approval or rejection of their borrow requests, improving communication and transparency.

Description

The Smart Library system is built on several interconnected ServiceNow components. Together they form a complete, automated library management solution. Below is a detailed description of each component and its role in the system.

Roles

Student and librarian roles manage access permissions on request of books. Role-based access control is a foundational security principle that ensures users can only perform actions they are authorized to carry out.

- Student Role: Can create new borrow requests, view the book catalog, and check the status of their own requests. Cannot modify book records or approve requests.
- Librarian Role: Has full visibility into all borrow requests and can approve or reject them. Can create, update, and delete book records in the catalog. Responsible for managing the overall library inventory.

Tables

Book and Borrow Request tables store catalog and transaction data. Two custom tables are at the heart of the system:

- Book Table (u_book): Stores the library catalog with fields for ISBN, Title, Author, and Status (Available/Issued). Each record represents a physical book in the library.
- Borrow Request Table (u_borrow_request): Records every borrowing transaction. Fields include the requesting student, the book being requested, request status, return date, and approval outcome.

Qualifier

Ensures only available books can be requested. The Reference Qualifier on the Book field in the Borrow Request table filters the book lookup to show only records where Status = Available. This prevents students

from

requesting books that are already issued to another borrower, maintaining data integrity automatically without requiring manual checks.

Flow

Automates approval workflows and updates book status. The Borrow Request Approval Flow is the core automation engine of the system. It listens for new borrow requests and orchestrates the entire approval process end-to-end, including conditional logic for both approved and rejected outcomes.

Policies

Lock or enforce field behavior after approval. UI Policies are client-side rules that dynamically control form fields based on conditions. In this system, UI Policies ensure that the Return Date field becomes mandatory whenever the borrow request status is set to Issued, preventing incomplete records.

Access Control Rules (ACLs)

Enforce role-based permissions for read, write, and delete operations. ACLs are server-side security rules that control which users can perform which operations on which records. They work in conjunction with UI Policies to provide defense-in-depth security across both the client and server layers.

Reports

Generate insights such as Most Borrowed Books. The reporting module provides administrators and librarians with data visualizations that reveal which books are in highest demand, how many requests are pending or approved, and overall library utilization trends.

Project Demonstration

The Smart Library system was developed across seven structured phases. Each phase built upon the previous one, creating a layered and cohesive system.

Milestone 1: Create Roles

Roles are the foundation of the system's security model. They define what each type of user can see and do within the platform. Without properly defined roles, there is no way to enforce access restrictions or separate student and librarian capabilities.

To create a role, provide a unique name and a clear description explaining the purpose of the role, then save the record. Once created, roles are assigned to individual user accounts through the User Administration module.

Steps:

- Navigate to User Administration → Roles.
- Create roles: librarian and student.
- Assign roles to users as required.
- Verify role assignments by logging in as a test user to confirm access restrictions are enforced.

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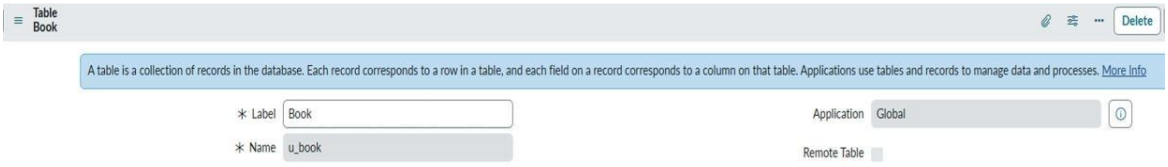
Milestone 2: Table Creation

Custom tables are the data backbone of the Smart Library system. ServiceNow's platform allows administrators to define new tables with custom fields, data types, and relationships without writing any database code. The Book and Borrow Request tables were designed to capture all information needed to manage the library's catalog and transactions.

Steps:

- Navigate to System Definition → Tables.
- Create the Book table (u_book) with fields: ISBN (String, max 20), Title (String, max 100, display=true), Author (String, max 100), Status (Choice: Available/Issued, default=Available), Student (Reference to Borrow Request).
- Create the Borrow Request table (u_borrow_request) with fields: Status (Choice: Requested/Approved/Rejected/Issued), Return Date (String, max 100), Book (Reference to Book table).
- Save both tables and verify they appear in the application navigator.

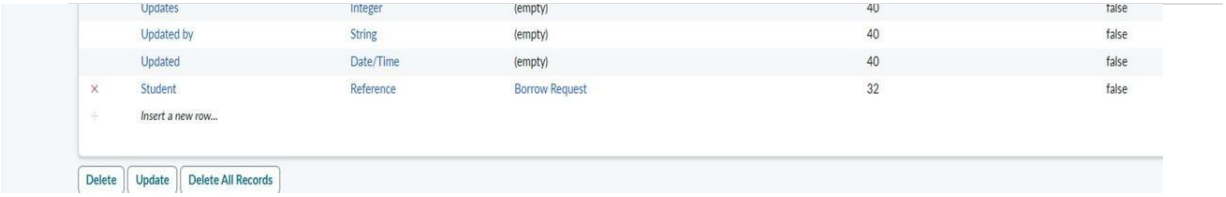


Book table column configuration (Part 1)

Figure 2.2 – Book table column configuration (Part 2)

×	ISBN	String	(empty)	20		false
×	Title	String	(empty)	100		true
×	Status	Choice	(empty)	40	Available	false
×	Author	String	(empty)	100		false
	Sys ID	Sys ID (GUID)	(empty)	32		false
	Created by	String	(empty)	40		false
	Created	Date/Time	(empty)	40		false

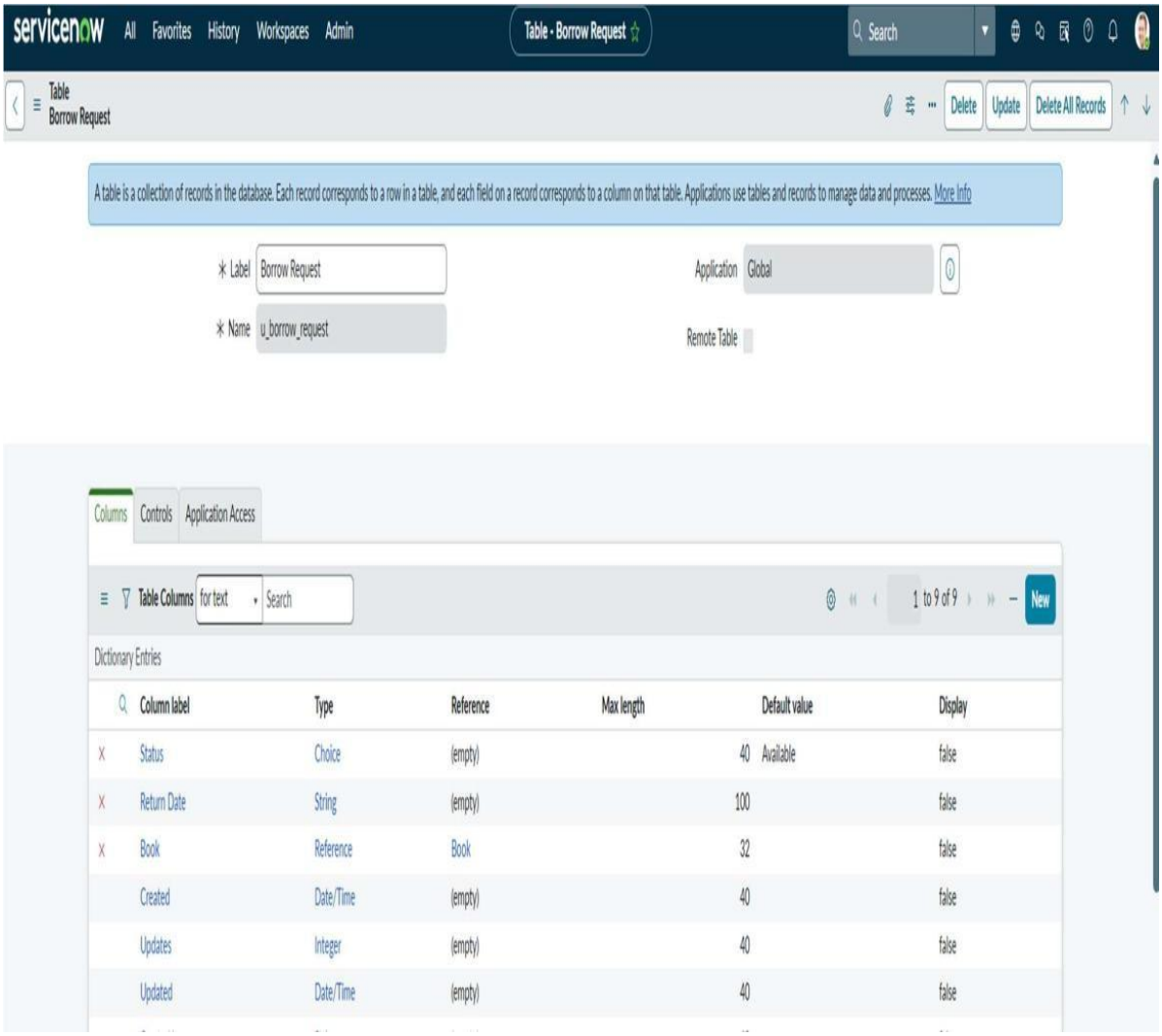
Book table column configuration (Part 3)



	Updates	Integer	(empty)	40	false
	Updated by	String	(empty)	40	false
	Updated	Date/Time	(empty)	40	false
×	Student	Reference	Borrow Request	32	false
	Insert a new row...				

Delete | Update | Delete All Records

Bo



servicenow All Favorites History Workspaces Admin | Table - Borrow Request | Search

Table Borrow Request | Delete | Update | Delete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info

* Label: Borrow Request
* Name: u_borrow_request
Application: Global
Remote Table:

Columns | Controls | Application Access

Table Columns for text Search | 1 to 9 of 9 | New

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
X	Status	Choice	(empty)	40	Available	false
X	Return Date	String	(empty)	100		false
X	Book	Reference	Book	32		false
	Created	Date/Time	(empty)	40		false
	Updates	Integer	(empty)	40		false
	Updated	Date/Time	(empty)	40		false

ok table column configuration (Part 4)

Created by	String	(empty)	40	false
Updated by	String	(empty)	40	false
Sys ID	Sys ID (GUID)	(empty)	32	false
Insert a new row...				

Delete | Update | Delete All Records

Related Links

Milestone 3: Configuring Table

After creating the tables, the Book field in the Borrow Request table was configured with a Reference Qualifier to enforce business rules at the data entry level. This configuration ensures that students can only select books that are currently available when creating a new borrow request.

Configure the Borrow Request table's Book field to reference only records where Status = Available. This is achieved by setting a Reference Qualifier condition on the Book reference field. The qualifier filters the lookup dynamically, so only books with Status = Available appear in the selection dropdown.

This approach is superior to manual validation because it prevents invalid data at the point of entry, rather than catching errors after the fact. It also improves the student's experience by presenting only valid options, reducing confusion.

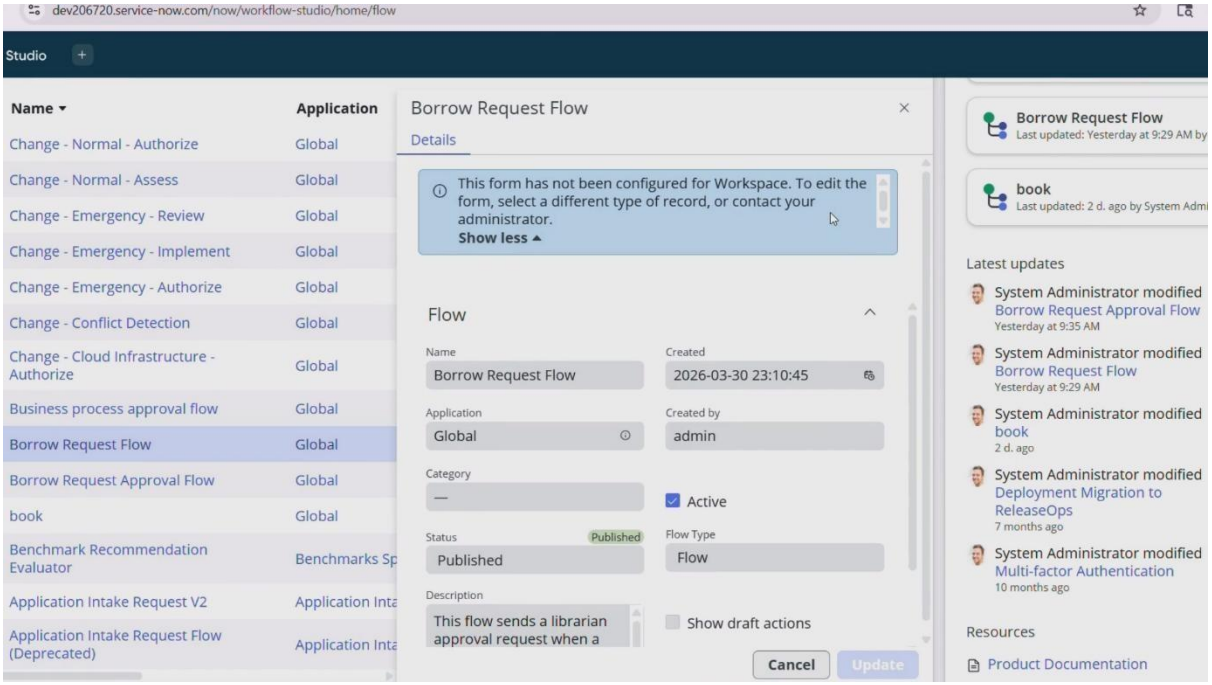
Milestone 4: Creating Flow Designer

The Flow Designer is ServiceNow's no-code/low-code automation tool. The Borrow Request Approval Flow is the central automation that drives the entire library request lifecycle. It eliminates the need for librarians to manually check for new requests, track approvals, and send notifications.

Flow Configuration:

- Create a flow named: Borrow Request Approval Flow.
- Trigger: Borrow Request created with Status = Requested — fires automatically whenever a student submits a new request.
- Action 1: Ask For Approval on Borrow Request — sends an approval task to the librarian.
- Action 2 (If Approved): Update Borrow Request Record — sets Status = Approved.
- Action 3 (If Approved): Update Book Record — sets Book Status = Issued, preventing others from requesting the same book.
- Action 4 (If Rejected): Update Borrow Request Record — sets Status = Rejected.
- Action 5: Update Borrow Request Record — final status update after all conditional branches.
- Action 6: Update Book Record — synchronizes book status with the final request outcome.
- Action 7: Update Borrow Request Record — records the final state for reporting purposes.
- Action 8: Send Email — dispatches an automated email notification to the student informing them of the approval or rejection outcome.

The flow's conditional branching (If Approved / If Rejected) ensures that each possible outcome is handled correctly and automatically, with no manual steps required from the librarian beyond clicking Approve or Reject on the approval task.



Milestone 5: Creating UI Policies

UI Policies control the behavior of form fields dynamically, based on conditions evaluated in real time as users interact with forms. They enhance the user experience and enforce data quality rules at the front-end layer, complementing the server-side ACLs.

Configuration:

- Navigate to System UI → UI Policies.
- Create a UI Policy for the Borrow Request table.
- Condition: Status = Issued.
- Action: Make the Return Date field mandatory when the condition is met.
- Enable 'On Load' so the policy applies when the form first loads, not just when the status field changes.
- This ensures a return date is always recorded when a book is issued, supporting accurate tracking and overdue management.

Milestone 6: Creating Access Control List

Access Control Lists (ACLs) are server-side security rules that define which users can perform which operations on records and fields in ServiceNow. They provide the critical security layer that prevents unauthorized data access or modification, regardless of what is shown or hidden in the UI.

Configuration:

- Navigate to Security → Access Control (ACL).
- Configure ACLs for the Book table (u_book).
- Students: read only — can view the catalog but cannot modify book records.
- Librarians: read, write, delete — full control over book records and borrow requests.
- Configure separate ACLs for both the Book table and the Borrow Request table.
- Test each ACL by logging in as a student and librarian to verify enforcement.

The combination of UI Policies (front-end) and ACLs (back-end) creates a defense-in-depth security model. Even if a user attempts to bypass the UI, the ACLs enforce restrictions at the platform level.

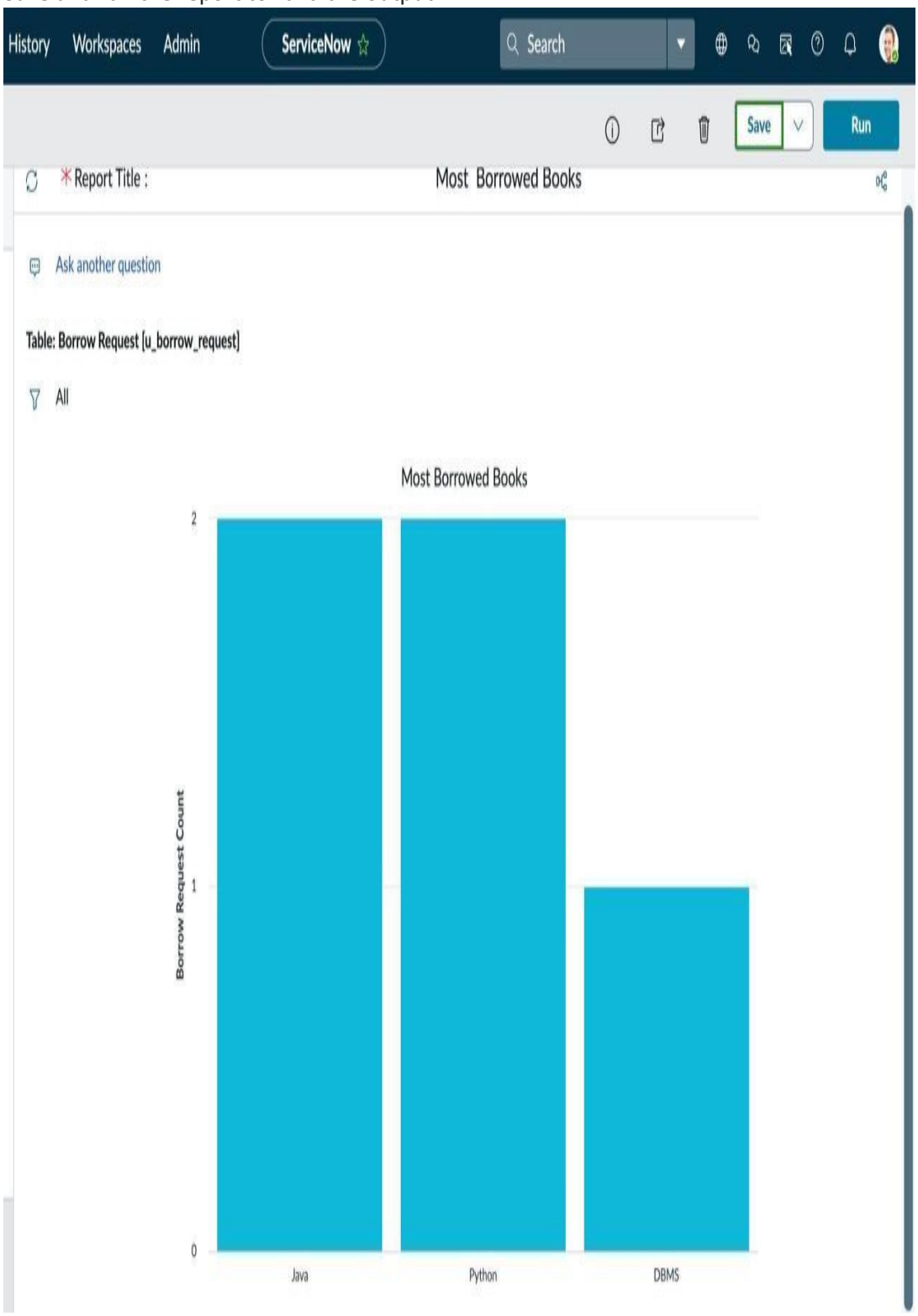
Milestone 7: Create Report

Reports transform raw data stored in the Borrow Request and Book tables into actionable insights. The Most Borrowed Books report gives administrators a clear view of which titles are in highest demand, enabling informed decisions about library acquisitions and inventory management.

Configuration:

- Navigate to Reports → Create New.
- Name the report: Most Borrowed Books.
- Source table: Borrow Request (u_borrow_request).
- Report type: Bar Chart — visually displays relative borrowing frequency per book.
- Group by: Book field — each bar represents one title.
- Aggregate by: Count — counts the number of borrow requests per book.
- Filter condition: Status = Approved — counts only successfully issued books.
- Share settings: Visible to Everyone, requires librarian or student role to view.

– Save and run the report to valid the output.



Project Explanation

The project integrates multiple ServiceNow features to build a functional library management system. Each component plays a distinct but complementary role in delivering the complete solution:

- Roles ensure secure access — by defining librarian and student roles, the system enforces the principle of least privilege, ensuring users can only perform actions relevant to their responsibilities.
- Tables store structured data — the Book and Borrow Request tables provide a clean, relational data model that supports all workflows and reporting requirements.
- Reference qualifiers maintain integrity — filtering available books in real time prevents conflicting requests and ensures data consistency.
- Flow Designer automates approvals and notifications — the end-to-end approval flow removes manual effort, speeds up processing, and keeps students informed throughout the process.
- UI Policies enforce consistency — dynamic form rules guide users correctly and prevent incomplete or invalid data entry.
- ACLs enforce security — server-side access controls ensure that the system remains secure regardless of how users interact with it.
- Reporting provides valuable insights — the Most Borrowed Books report and related analytics help administrators manage the library proactively.

Together, these components deliver a system that is not only functional and efficient, but also secure, scalable, and easy to maintain. The use of ServiceNow's native platform capabilities means there is no custom code to maintain — all logic lives within the platform's declarative configuration layers.

Implementation

For this project, we built the Smart Library Request Workflow entirely on ServiceNow. The setup allows students to raise book requests online, which then go directly to the librarian for approval. Once approved, the book is marked as issued and the system keeps track of it until it’s returned.

While building the system, a few important decisions were made to make sure everything worked correctly and reliably:

- The Reference Qualifier was set at the dictionary level for the Book field, ensuring it applies consistently across all forms and list views.
- The Flow Designer trigger was configured to fire on both creation and update of Borrow Request records, ensuring that status changes are always processed correctly.

- ACLs were tested using dedicated test user accounts with only the student or librarian role, verifying that access restrictions were properly enforced.
- The email notification template was customized to include the book title, student name, and approval outcome, making communications clear and actionable.
- The Most Borrowed Books report was shared with the Everyone audience but restricted by role, so only authenticated library users can view it.

Results

Once the system was up and running, it made a real difference. Book requests that used to require backand-forth communication were now handled automatically, and the entire process became much smoother for both students and librarians.

Here's what we observed after deploying the system:

- All book requests are now processed through a standardized, automated workflow — eliminating ad-hoc manual handling.
- Librarians receive structured approval tasks in their ServiceNow inbox, reducing the chance of requests being missed or delayed.
- Students receive immediate email confirmation when their requests are approved or rejected, improving communication.
- Book availability is always accurately reflected in the system — once issued, a book no longer appears in the available selection.
- The Most Borrowed Books report showed that Java and Python were the most requested titles, which gives the library useful data when deciding what new books to add.
- Return dates are consistently recorded for all issued books, supporting future implementation of overdue tracking.

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Conclusion

Overall, this project turned out to be a pretty satisfying build. The Smart Library system does exactly what it was supposed to — it simplifies the book borrowing process, takes the manual work off the librarian's plate, and keeps students informed at every step. What started as a straightforward workflow problem ended up being a good example of how ServiceNow can be used for more than just IT tasks.

One thing that stood out while working on this was how much you can build on ServiceNow without writing a single line of custom code. Everything — from the approval flow to security rules to reports — was done using the platform's built-in tools. That makes it a lot easier to maintain and scale in the future.

The seven phases we followed — Roles, Tables, Table Configuration, Flow Designer, UI Policies, ACLs, and Reports — form a solid blueprint that could be reused for other similar projects. If someone wanted to build a similar system for equipment loans or room bookings, the same approach would work just as well.

Future Scopes

The system works well as it is, but there's definitely room to grow. Here are a few things that could make it even better down the line:

- Scheduled Jobs: Automate recurring tasks such as overdue reminders — a daily scheduled job could check for borrow requests where the Return Date has passed and the book has not been returned, automatically notifying the student and flagging the record for librarian review.
- Integration with Digital Resources: Extend to e-books and online catalogs — adding support for digital titles would allow the same approval workflow to manage access to electronic resources, expanding the system

. Enhanced Notifications: SMS or mobile app alerts for approvals and returns — integrating with ServiceNow's notification channels or third-party SMS providers would ensure students receive timely updates even when not actively checking their email.

- Advanced Analytics: Track borrowing trends, user activity, and resource utilization — building additional reports and performance dashboards would give library administrators deeper insights into usage patterns over time.
- Scalability: Extend framework to other academic services such as reservations or asset tracking — the same ServiceNow table and workflow model could be applied to equipment loans, study room reservations, or other institutional resource management needs.
- Mobile-Friendly Portal: Create a ServiceNow Service Portal page for the library — this would give students a consumer-grade interface for browsing the catalog and submitting requests directly from their phones.

References

Sources used: ServiceNow official documentation, Flow Designer guides, and online tutorials related to IT Service Management (ITSM) and workflow automation.

- ServiceNow Documentation – Flow Designer: https://docs.servicenow.com/bundle/flow-designer
- ServiceNow Documentation – Access Control Rules: https://docs.servicenow.com/acl
- ServiceNow Documentation – UI Policies: https://docs.servicenow.com/ui-policies
- ServiceNow Community – Library Management Use Cases and Best Practices
- IT Service Management (ITSM) Workflow Automation – Online Tutorials and Guides